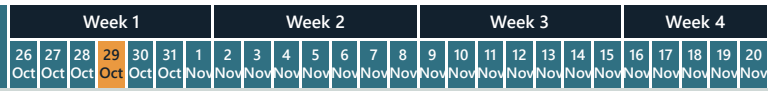
PROJECT	INSTRUCTOR	COURSE	DATE
Biliran Resort & Tour Booking System	**Engr. Eunel O. Peñaranda**	**Parallel and Distributed System**	**October 26 – November 20**

Team: Member A — central node & integration · Member B — island node · Member C — mobile app · Member D — consoles, reports & testing

Done · Ongoing · Planned · Today (29 Oct)

No.	Tasks and action items	Task owner	Start	Due	Days	Remarks / annotations	% done	Week 1 (26 Oct – 1 Nov)	Week 2 (2 Nov – 8 Nov)	Week 3 (9 Nov – 15 Nov)	Week 4 (16 Nov – 20 Nov)
PHASE 1: Project Planning and Requirements											
1.1	Select project and define problem statement, objectives, and scope	Member A + Member B	10/26	10/26	1	Problem: online and walk-in sales of island resort/tour packages share the same rooms, seats and boats, so the last unit can be sold twice; island links to the mainland are unreliable. Objective: book packages online and face-to-face from one inventory with no double booking, and keep islands working offline. Scope: prototype, 3 island nodes + 1 central node, simulated payment, fictional businesses. **Evidence:** Design doc §1–3	**100%** Done				
1.1.1	Assign roles, responsibilities, and individual deliverables	Member A	10/26	10/26	1	Member A: central node, booking saga, central admin panel, Supabase, gRPC contract, integration. Member B: island node (inventory, packages, hold/confirm, accounts, sync). Member C: customer mobile app (Expo). Member D: node console UI, parallel reports, load tests, benchmark charts. Every member explains the overall design and their own module. **Evidence:** Design doc §18, §5.1	**100%** Done				
1.2	Identify the parallel and distributed components of the proposed system	Member A + Member B	10/26	10/28	2	Distributed: each island node owns its data (SQLite) and talks to central over gRPC; outbox sync with retries and de-duplication; islands keep working offline (partition). Parallel: central queries all islands concurrently (fan-out); concurrent bookings made safe with atomic guarded updates; reports computed with worker threads. **Evidence:** Design doc §4.1, §12, §13	**100%** Done				
PHASE 2: System Framework and Technical Design											
2.1	Prepare system architecture, workflow, and framework	Member A + Member B	10/27	10/29	2	Architecture: 4 laptops on a hotspot — central (API, saga, admin panel, Supabase) + 3 island nodes (server, SQLite, console); phones run the app. IPO conceptual framework, 7 flowcharts (online booking, walk-in, arrival, mark unavailable, catalog change, registration/accounts), sequence and state diagrams. Revised to v2 after adviser feedback; awaiting approval. **Evidence:** Design doc §4, §5, §8–10; Framework-and-Flowcharts.pdf	**90%** Ongoing				
2.2	Define modules, interfaces, data flow, and technology stack	Member A + Member B	10/28	10/30	2	Modules M1–M11 (app, central API, orchestrator, admin panel, central sync, island booking engine, catalog/accounts, sync agent, node console, report workers, test tools). Interfaces: HTTP/JSON (app, consoles, admin), gRPC IslandService/CentralService. Stack: Expo (React Native, TypeScript), Node.js + TypeScript, gRPC, SQLite, Supabase PostgreSQL, worker_threads. **Evidence:** Design doc §5.1–5.3, §11, §15; proto/booking.proto	**90%** Ongoing				

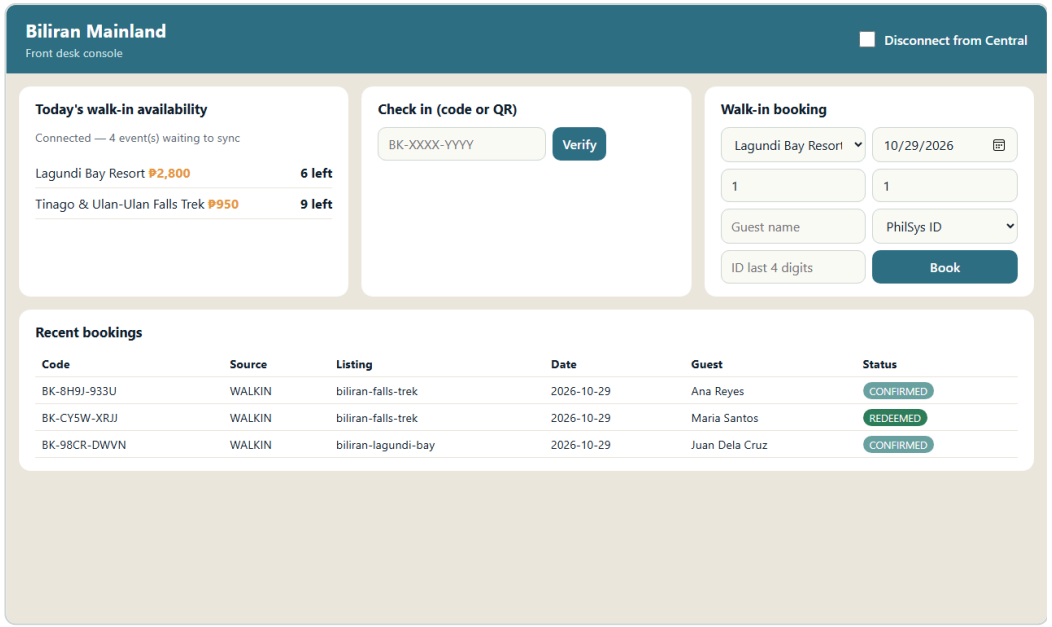
No.	Tasks and action items	Task owner	Start	Due	Days	Remarks / annotations	% done
2.3	Submit and defend the initial system framework for instructor approval	Whole team	10/30	10/30	1	Prepared: framework document (architecture diagram, workflow, module list, interfaces, data flow, technology stack) and short review slides. Defense scheduled 30 Oct; approval pending. **Evidence:** Upload: Task-2.3-Initial-System-Framework.pdf (framework + slides, one file)	**50%** Ongoing
PHASE 3: Core System Development							
3.1	Develop the basic functional system (baseline version)	Member A + Member B	10/29	11/3	5	Baseline v1 runs: 3 island nodes + central over gRPC, parallel search, walk-in booking from the island inventory, check-in once only, outbox sync to central, Supabase online booking. Rework to v2 (packages, hold → pay → confirm, no tickets) started. **Evidence:** Tag baseline-v1-2026-10-29; screenshot attached	**30%** Ongoing
3.2	Implement core features and individual modules	Members B, C, D	11/2	11/6	4	Planned: package catalog and per-guest items, mark-unavailable (blocked units), arrival confirmation by reference/name, node admin pages, receptionist accounts and local login, customer app screens. Starts once the v2 baseline is usable.	**0%** Planned
PHASE 4: Parallel and Distributed Implementation							
4.1	Implement parallel processing or concurrent task execution	Member D + Member A	11/3	11/9	6	Done in prototype: concurrent fan-out search (measured 2.93× vs sequential). To do: occupancy/sales reports split across 1/2/4/8 worker threads, compared with a sequential run; record test conditions. **Evidence:** loadtest/search.ts	**15%** Ongoing
4.2	Implement distributed communication, coordination, or remote processing	Member A + Member B	11/4	11/10	6	Done in prototype: gRPC between nodes with deadlines, idempotency keys, outbox with retries and de-duplication by event id, partition switch. To do for v2: HoldBooking/ConfirmBooking/ReleaseHold, PullDirectory, reconnect handling. **Evidence:** shared/rpc.ts, island/server.ts	**30%** Ongoing
4.3	Integrate parallel and distributed components	Whole team	11/9	11/11	2	Demonstrate one complete booking: app → central → island hold → payment → confirm → arrival at the desk → sync to central → visible in admin reports.	**0%** Planned
PHASE 5: Integration and Testing							
5.1	Conduct functional, integration, and error-handling tests	Member C + Member D	11/10	11/12	2	Prototype end-to-end checks pass (10/10). Test cases to attach: valid booking, sold out, duplicate request, payment failure, hold expiry, double arrival, island offline, reconnect and sync; expected vs actual, pass/fail.	**10%** Ongoing
5.2	Collect performance evidence and explain parallel/distributed behavior	Member D	11/11	11/13	2	Preliminary: search 2.93× faster in parallel; 30 concurrent requests for 6 rooms → 0 oversold. Final: 100 / 500 / 1,000 requests on the hotspot, report speedup with 1/2/4/8 threads, compared with Amdahl's law. No speedup claimed without measurements.	**15%** Ongoing
PHASE 6: Documentation and Evidence							



No.	Tasks and action items	Task owner	Start	Due	Days	Remarks / annotations	% done	Week 1							Week 2							Week 3							Week 4				
								26 Oct	27 Oct	28 Oct	29 Oct	30 Oct	31 Oct	1 Nov	2 Nov	3 Nov	4 Nov	5 Nov	6 Nov	7 Nov	8 Nov	9 Nov	10 Nov	11 Nov	12 Nov	13 Nov	14 Nov	15 Nov	16 Nov	17 Nov	18 Nov	19 Nov	20 Nov
6.1	Technical documentation and individual contribution records	Member D	11/13	11/16	3	Existing: README (setup), DESIGN.md, 34-page design document, commented code. To add: testing results, limitations, individual contribution records. **Evidence:** README.md, DESIGN.md	**20%** Ongoing																													
PHASE 7: Final Validation and Defense Preparation																																	
7.1	Prepare slides, architecture visuals, and demonstration sequence	Member C	11/13	11/16	3	Final slides: problem, architecture, parallel/distributed workflow, test evidence, limitations, live demo order (register island → publish package → book online → walk-in race → island offline → benchmarks).	**0%** Planned																													
7.2	Conduct mock defense and individual questioning	Whole team	11/16	11/17	1	Timed mock defense; each member answers questions on their own module.	**0%** Planned																												
7.3	Fix critical bugs and verify the final system version	Member B	11/16	11/18	2	Fix critical defects, rerun regression tests, record remaining known issues.	**0%** Planned																												
7.4	Submit final project package and prepare backup demonstration	Member A + Member D	11/18	11/18	1	Report, source code, README/setup guide, dependency list, test evidence, contribution record, slides, recorded backup demo.	**0%** Planned																												
PHASE 8: Final Defense																																	
8.1	Final project defense and live demonstration	Whole team	11/19	11/20	1	Live demo on four laptops and phones: online and walk-in booking, race for the last room, island offline and recovery, measured speedups, failure handling.	**0%** Planned																												

Evidence attached (status as of 29 October 2026)

Task	File / evidence (repository github.com/Abylon10/BookingSystem)
1.1, 1.1.1, 1.2	**docs/System-Design-and-Initial-Considerations.pdf** §1 Introduction, §2 Objectives, §3 Scope and limitations, §6 Roles, §4.1 and §13 parallel/distributed components; §18 team roles
2.1	Same document §4 conceptual framework (IPO), §5 architecture diagram, §8 flowcharts, §9 sequence diagrams; short version **docs/Framework-and-Flowcharts.pdf**
2.2	Same document §5.1 module list, §5.2 interfaces, §5.3 data flow, §11 data model (ERD), §15 technology stack; gRPC contract **proto/booking.proto**
2.3	**docs/Task-2.3-Initial-System-Framework.pdf** — single upload: framework document, architecture diagram, workflow, module list, technology stack, and the 11 short review slides (also separately: docs/Framework-Review-Slides.pdf)
3.1	Baseline prototype, source version git tag **baseline-v1-2026-10-29**. Screenshot: **docs/evidence/2026-10-29-baseline-island-console.png** (right)
4.1, 4.2, 5.2	Preliminary measurements from the prototype (local laptop, 150 ms simulated link): parallel search 2.93× faster than sequential (180 ms vs 527 ms, 10 runs each); 30 concurrent bookings for 6 rooms → exactly 6 booked, 0 oversold; parallel double check-in → exactly one success
6.1	**README.md** (setup), **DESIGN.md** (v2 design), every code block commented



Baseline (task 3.1): Biliran island node console running the v1 prototype — walk-in availability, three walk-in bookings, one guest checked in (REDEEMED), and four events waiting in the outbox to sync to central. Captured 29 Oct 2026. The v1 prototype predates the adviser's v2 changes (packages, no tickets); it is being reworked to v2 under tasks 3.1–3.2.

Notes

- Percentages are honest estimates as of 29 Oct 2026. Measurements in 4.1/4.2/5.2 are preliminary (prototype on one laptop); final evidence will be measured on the four-laptop hotspot setup with 100 / 500 / 1,000 requests.
- Design was revised to v2 on 29 Oct after adviser feedback: no inventory split, receptionist marks unavailable services, no tickets, package booking, central admin + node admins + receptionists.
- Known open items: framework approval (2.3), v2 rework of the island data model, hotspot test with all 8 devices.